

A Generalizable Multimodal AI Framework for Biomarker-Based Assessment of Cognitive and Mental Health

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Abstract—Multimodal artificial intelligence (AI) offers significant potential for objective assessment of cognitive and mental health, yet existing systems remain limited by modality imbalance, inconsistent evaluation practices, and insufficient interpretability. This paper proposes a generalizable multimodal AI framework that integrates speech-based, behavioral, physiological, and structural imaging biomarkers for comprehensive mental- and cognitive-health analysis. Using the OASIS MRI dataset as a core benchmark, alongside speech- and behavioral datasets such as DAIC-WOZ, the framework incorporates hybrid machine-learning and deep-learning pipelines, including CNNs, Transformers, and pre-trained models such as Wav2Vec2.0 and BERT. Dynamic fusion strategies and temporal modeling enable robust prediction of cognitive decline and mental-health states, while explainable AI techniques (SHAP, LIME) provide the transparency required for clinical and research adoption. The proposed system emphasizes reproducibility, interpretability, and cross-dataset generalizability, offering a scalable foundation for biomarker-based cognitive and mental-health evaluation.

Keywords—Multimodal AI, digital biomarkers, OASIS dataset, DAIC-WOZ, cognitive health, mental health, explainable AI, SHAP, LIME, deep learning.

I. INTRODUCTION

Cognitive and mental-health disorders such as Alzheimer’s Disease (AD), mild cognitive impairment (MCI), depression, and anxiety continue to pose significant clinical and societal burdens. Traditional diagnostic approaches rely heavily on subjective self-reports, infrequent clinical assessments, and resource-intensive neuroimaging procedures, limiting their ability to support early detection or continuous monitoring. In contrast, recent advances in multimodal artificial intelligence (AI) have enabled the extraction of objective biomarkers from speech, behavioural signals, physiological data, and medical imaging, offering new pathways for scalable and data-driven assessment.

Multimodal systems consistently outperform unimodal approaches by integrating heterogeneous yet complementary sources of information. Speech biomarkers have shown strong potential for cognitive-impairment screening and depression detection, while behavioural and physiological signals captured from remote settings further enhance mental-health monitoring. Structural MRI imaging remains a key modality in neurodegenerative research, with the OASIS dataset serving as a widely adopted benchmark for AD classification and progression modeling. Despite these advances, existing models struggle with cross-population

generalizability, modality imbalance, and limited interpretability—barriers that must be addressed before real-world adoption.

To overcome these challenges, this research proposes a generalizable multimodal AI framework that unifies speech, behavioural, physiological, and neuroimaging biomarkers into a single, scalable system. The framework employs hybrid machine-learning and deep-learning techniques—including CNNs, Transformers, and pre-trained models such as Wav2Vec2.0 and BERT—supported by robust preprocessing and temporal modeling pipelines. Explainable AI methods (SHAP, LIME) are integrated to ensure transparency and clinical trust. By combining open-access datasets such as OASIS and DAIC-WOZ with advanced fusion and interpretability techniques, this work aims to establish a reproducible and ethically aligned foundation for cognitive and mental-health assessment.

II. LITERATURE REVIEW

1. Biomarker Validation and Scope Across Health Domains

A robust multimodal framework requires validated biomarkers across cognitive and mental health domains [4], [15].

1.1 Cognitive Health Biomarkers

Non-invasive cognitive screening methods increasingly rely on speech biomarkers derived from spontaneous or simulated interactions [6]. For instance, temporal linguistic cues—such as pause length and position—serve as strong dementia-sensitive features whose interpretability can be enhanced through adversarial learning [12].

High-accuracy convolutional models trained on the OASIS MRI dataset have achieved up to 98.8% accuracy in classifying Alzheimer’s Disease (AD) stages [8], establishing a crucial single-modality benchmark. To capture chronic disease progression, longitudinal modeling approaches such as autoregressive (AR) models are employed to track temporal variation in both MRI-derived and neuropsychological measures [5].

1.2 Mental Health Biomarkers

In mental health assessment, multimodal detection fuses facial, vocal, linguistic, and physiological signals collected remotely [10], [15]. Combining objective sensor data, such as ECG and sleep tracking, with self-reported questionnaires

has achieved F1-scores near 94% in stress prediction and early mental health monitoring [4].

Physiological signals like EEG and eye-movement patterns provide complementary features critical for emotion and stress recognition, and their integration benefits from adversarial multitask fusion to preserve modality-specific information [17]. Even low-complexity acoustic features processed by 1D-CNNs demonstrate cross-linguistic robustness, maintaining accuracy across English and Chinese datasets [14]. A key benchmark for generalizability is the ability to simultaneously separate overlapping conditions such as depression, Parkinson’s Disease, and phonation disorders [7], [13].

2. Generalizability and Advanced Fusion Architectures

Ensuring universal model robustness demands addressing the complexities of multimodal fusion and cross-domain variability [2], [16].

2.1 Dynamic and Personalized Fusion

Modern architectures employ Adaptive Gated Fusion (AGF) and cross-modal attention to dynamically adjust each modality’s contribution [18]. This adaptability allows models to integrate personalized factors—such as personality traits—to improve diagnostic relevance [18]. For modeling nonlinear dependencies among modalities, Cross-Modal Graph Fusion (CGF) with attention aggregation provides a flexible mechanism for capturing intersignal relationships [16].

When combining distinct physiological signals (EEG, eye movement), multitask adversarial training explicitly preserves complementary modality-specific information [17], enhancing fusion efficiency and resilience.

2.2 Temporal and Structural Modeling

Accurately modeling longitudinal progression remains central to chronic condition tracking [5]. Simpler trimodal fusion methods (audio, video, text) risk information loss [15], but optimized hybrid models such as CNN–SVM classifiers demonstrate efficiency in resource-constrained environments, making them suitable for embedded systems [9].

Furthermore, specialized LSTM models combined with multimodal learning analytics prove effective for real-time psychological state prediction in non-clinical contexts, successfully managing challenges related to heterogeneous data sampling rates [3]. This advantage of combining modalities is demonstrated by models achieving F1-scores near 98% in depression detection using feature-level fusion of audio, video, and text with simple machine-learning classifiers [19].

2.3 Cross-Domain Robustness

Cross-lingual validation studies confirm that acoustic biomarkers generalize well between languages (e.g., English → Greek) [2]. This demonstrates that certain prosodic and acoustic features remain stable across linguistic contexts [14]. However, continued research into deep feature representations is needed to strengthen predictive performance and ensure consistent results across geographically and linguistically diverse populations [6], [14].

3. Ethical Mandates and Future Research Directions

For successful clinical deployment, frameworks must prioritize interpretability, data ethics, and standardized evaluation [11], [4], [18].

3.1 Interpretability and Explainability (XAI)

Clinical acceptance depends on transparent and explainable AI systems [13], [8]. Integrating explainable AI (XAI) tools—such as SHAP and LIME—can clarify how multimodal inputs influence diagnostic outcomes [13], [12]. These mechanisms should explain not only individual biomarker relevance but also temporal progression predictions.

3.2 Ethical and Temporal Gaps

Continuous digital monitoring raises serious data privacy and consent concerns [4], [18]. To ensure scalability while preserving user confidentiality, federated learning architectures enable decentralized, privacy-preserving model training [1]. Methodologically, future research must address incomplete longitudinal data and avoid simplistic imputation strategies that weaken predictive stability [5].

3.3 Standardization and Validation

Despite rapid technical advances, the field lacks uniform evaluation protocols [11]. This absence hinders reproducibility and limits meaningful cross-study comparison. Establishing standardized validation frameworks and reporting guidelines is essential for consistent clinical translation of multimodal AI systems.

4. Figures and Tables

Table I presents multimodal fusion approaches that integrate audio, visual, physiological, behavioral, and textual biomarkers using advanced fusion architectures.

Table II focuses on speech-only methods leveraging acoustic and prosodic features for detecting depression, cognitive decline, and neurological disorders.

Table III highlights imaging and cognitive-biomarker studies centered on MRI and neuropsychological modeling. Together, these tables outline current strengths and gaps across modalities and motivate the need for a unified, generalizable multimodal framework.

TABLE I. MULTIMODAL FUSION APPROACHES

Multimodal Fusion Methods for Mental-Health and Cognitive Assessment				
Ref	Approach	Advantages	Disadvantages	Gaps
Pradnya et al. (2025)	Multimodal depression detection using text, images, and MBTI traits with AGF + CMA fusion mechanisms.	Personalized predictions; dynamic modality weighting	Heavy dependence on noisy social-media data; requires MBTI extraction from third-party tools.	Need longitudinal data to model symptom progression; requires improved interpretability for clinical trust.
Yang et al. (2025)	Emotion recognition AAT for modality alignment + CGF for graph-based multimodal fusion across audio, visual, and text.	Models complex inter-modal relationships; high benchmark performance.	Computationally heavy; difficult interpretability.	Real-time deployment and cross-cultural generalization needed.
Jiang et al. (2024)	Multimodal depression/anxiety detection combining facial landmarks, speech prosody, linguistic features, and rPPG.	Supports remote, low-cost assessment; integrates diverse biomarkers for higher robustness.	Device inconsistencies affect rPPG and audio quality; temporal sparsity reduces stability.	Requires bias mitigation for different device types; needs improved temporal alignment techniques.
Govind et al. (2023)	Trimodal early fusion of audio, video, and text using PCA for dimensionality reduction + logistic regression.	High F1 performance using simple models; computationally efficient; avoids deep model overfitting.	Early fusion cannot dynamically weight modalities; deep models limited by small dataset.	Explore deep fusion and attention-based weighting; increase dataset size for generalization.
Kowalewski et al. (2023)	Audio + video multimodal depression detection using hand-crafted features and XGBoost.	Strong robustness to heterogeneous data; good performance on noisy samples.	Does not use textual features; limited temporal modeling.	Add linguistic/sentiment features; explore temporal attention or RNNs.
Liu et al. (2025)	LSTM architecture integrating visual, behavioral, and interaction-based analytics.	Handles heterogeneous sampling rates; supports real-time prediction.	Lower F1 for darker skin tones (bias); lacks physiological signals.	Needs federated learning to reduce demographic bias; add physiological context.
Indumathy et al. (2025)	Transformer-based fusion of ECG, sleep patterns, and questionnaire responses for teen stress/anxiety.	High F1; strong early detection; integrates physiological and behavioral signals.	Youth-only dataset limits generalization; ethical/privacy concerns.	Real-world deployment; privacy-aware data pipelines; expand beyond adolescent populations
Khan et al. (2025)	Transformer + Federated Learning using handwriting/drawing as biomarkers.	Strong privacy guarantees; lightweight for IoMT devices.	Narrow demographic; lacks longitudinal data.	Expand to diverse populations; validate long-term emotional trajectories; secure aggregation
Liu et al. (2023)	EEG + eye movement fused with multitask adversarial training + attention.	Preserves complementary physiological features; strong 94.8% accuracy; interpretable via attention.	Expensive sensors; subject-level variability; limited scalability.	Evaluate across broader populations; reduce hardware dependency; cross-dataset validation.

Abbreviations— AGF: Adaptive Gated Fusion; CMA: Cross-Modal Attention; AAT: Attention Aggregation Technique; CGF: Cross-Modal Graph Fusion; rPPG: Remote

TABLE II. SPEECH-ONLY BIOMARKER APPROACHES

Speech-Based Methods for Cognitive and Mental-Health Assessment				
Ref	Approach	Advantages	Disadvantages	Gaps
Kumar & France (2025)	PD detection using jitter, HNR, and DFA acoustic biomarkers with IF-XGBoost + SHAP explanations.	Achieves 96.61% accuracy; interpretable predictions via SHAP; uses clinically meaningful features.	Limited dataset size; only voice modality used.	Add multimodal features (gait/handwriting); validate clinically with larger samples.
Zhu et al. (2022)	Dementia detection using WavBERT pause-duration features with adversarial training.	Effective modeling of temporal speech pauses; supports interpretability through attention maps.	ASR misspellings reduce accuracy; unimodal system.	Large-scale pretraining to reduce ASR noise; incorporate linguistic context.
Pacheco - Lorenzo et al. (2024)	Voice-biomarker cognitive screening using smart speakers in naturalistic settings.	Ecological validity; non-invasive; uses conversational interactions.	Small, non-diverse sample; mixed conversation types lower consistency.	Multilingual validation; larger cross-cultural datasets.
Sztahó et al. (2018)	Speech-based multi-class classification using correlation of time-shifted features.	Novel approach for differentiating PD, depression, phonation disorders.	Confusion among phonation subtypes; limited robustness.	Add richer acoustic features; integrate sequential modeling.
Gheorghe et al. (2023)	1D-CNN with MFCC, APF, SPF features for depression detection.	92.10% accuracy; robust across languages; efficient model.	SMOTE duplicates information; single modality.	Evaluate cross-lingual transfer learning; integrate prosodic features.
Saidi et al. (2020)	Spectrogram-based CNN-SVM hybrid for depression detection.	Resource-efficient; suitable for embedded devices.	Lower accuracy compared to multimodal deep learning.	Develop embedded deployment; explore attention mechanisms.
Luz et al. (2023)	eGeMAPS acoustic features for multilingual AD detection (English→Greek).	Strong evidence of cross-lingual transfer; bias-controlled comparison.	Acoustic-only features limit predictive performance.	Identify optimal transferable acoustic features; integrate prosody + embeddings.

Abbreviations— MFCC: Mel-Frequency Cepstral Coefficients; APF: Amplitude Perturbation Features; SPF: Spectral Perturbation Features; IF-XGBoost: Interpretable Features-Extreme Gradient Boosting; SHAP: Shapley Additive Explanations; ASR: Automatic Speech Recognition; eGeMAPS: Extended Geneva Minimalistic Acoustic Parameter Set; PD: Parkinson's Disease.

TABLE III. IMAGING & COGNITIVE BIOMARKERS

Structural Imaging and Cognitive Biomarker Approaches				
Ref	Approach	Advantages	Disadvantages	Gaps
Minhas et al. (2018)	MRI + neuropsychological scores modeled with AR for MCI→AD progression.	Captures longitudinal biomarker evolution; multi-predictor models outperform single biomarkers.	Poor LOCF performance; short 3-year follow-up.	Improved imputation (e.g., deep imputation); longer longitudinal datasets.
Kumar & Nelson (2024)	CNN-based MRI classifier for AD stage identification on OASIS.	98.80% accuracy; strong imaging benchmark.	MRI costly; CNN lacks interpretability.	Add XAI (e.g., Grad-CAM/SHAP); improve early-stage classification.
Wolfert et al. (2022)	Evaluation frameworks for gesture generation in ECAs.	Establishes standardized evaluation practices applicable to clinical AI.	Not directly diagnostic; indirect relevance.	Develop clinical-grade evaluation standards for generalizable AI systems.

Abbreviations—MRI: Magnetic Resonance Imaging; AR: Autoregressive Modeling; LOCF: Last Observation Carried Forward; AD: Alzheimer’s Disease; MCI: Mild Cognitive Impairment; CNN: Convolutional Neural Network; ECA: Embodied Conversational Agent; XAI: Explainable Artificial Intelligence.

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